Communication modality influences neighborhood density

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1. Introduction

Languages adapt to minimize effort and maximize communicative success (Flemming, 2004; Lindblom & Maddeison, 1988). Written communication has increasingly replaced spoken language as the primary communication mode. This shift is significant because comprehension processes differ fundamentally: dense phonological neighborhoods create interference in sequential spoken word recognition (Marslen-Wilson, 1984; Luce & Pisoni, 1998), while parallel letter processing enables dense neighborhoods to facilitate visual recognition (Andrews, 1997; Yates, 2005). We tested whether languages that are used in the written modality evolve denser neighborhoods than those that are used in the spoken modality. This would also suggest that languages are currently changing to become more dense.

2. Methods

Study 1: Iterated learning (Kirby et al., 2008) in either spoken (15 chains) or written (15 chains) modality. Generation 1 named 12 novel objects. Subsequent generations learned and reproduced names, with their output becoming input to the following participant.

Study 2: Communication game in either spoken or written modality. 60 pairs communicated about novel objects using names that they invented.

3. Results

Contrary to hypothesis, in both studies the spoken condition had denser phonological neighborhoods than the written one (iterated: $p=0.008$; communication: $p=0.001$). Furthermore, neighborhood density reduced accuracy in the written ($p<2e-16$) but not the spoken modality ($p=0.52$), as reflected in an interaction between neighborhood density and modality ($p<2e-16$).

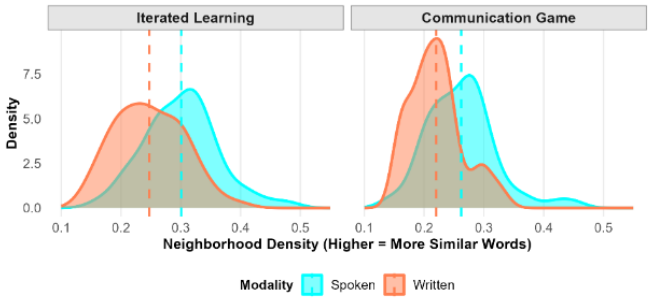

Figure 1. Neighborhood density by modality. Dashed lines show condition means.

4. Discussion

These findings demonstrate that modality influences neighborhood density, though in the opposite direction to the predicted one. Surprisingly, dense neighborhood had a more detrimental effect on comprehension in the written modality, and, accordingly, languages in the written modality evolved to have sparser neighborhoods. Reduced production demands in the written modality might have further enabled greater optimization. The convergent results across paradigms suggest the results are robust. These results suggest that language might be currently evolving to become phonologically sparser.

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6. References

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